Due Date

You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Monday, November 3, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Abstract Classes

This lab is an exercise in extending an abstract class to create a concrete class. Use the "TestGeometricObject" sample program (from the textbook, and available on *Moodle*) as a guide. This sample program contains source code for the abstract **GeometricObject** class, as well as the textbook examples for the **Circle** and **Rectangle** classes.

Programming Exercise

Design and code a new **Triangle** class that extends the abstract **GeometricObject** class. The **Triangle** class must contain:

- Three double data fields named **`side1`**, **`side2`**, and **`side3`** with default values **`1.0`** to denote the sides of the triangle.
- A no-arg constructor that creates a default triangle.
- A constructor that creates a triangle with the specified **side1**, **side2**, and **side3** values.
- Accessor methods ("getter" methods) for each of the three data fields.
- A method named **`getArea()`** that returns the area of this triangle.
- A method named **`getPerimeter()`** that returns the perimeter of this triangle.
- A method named **`toString()`** that returns a description of the triangle.

Write a test program that prompts the user to enter three sides of the triangle, a color, and a Boolean value to indicate whether the triangle is filled. The program should display the area and perimeter values, as well as the color and **true** or **false** to indicated whether it is filled.

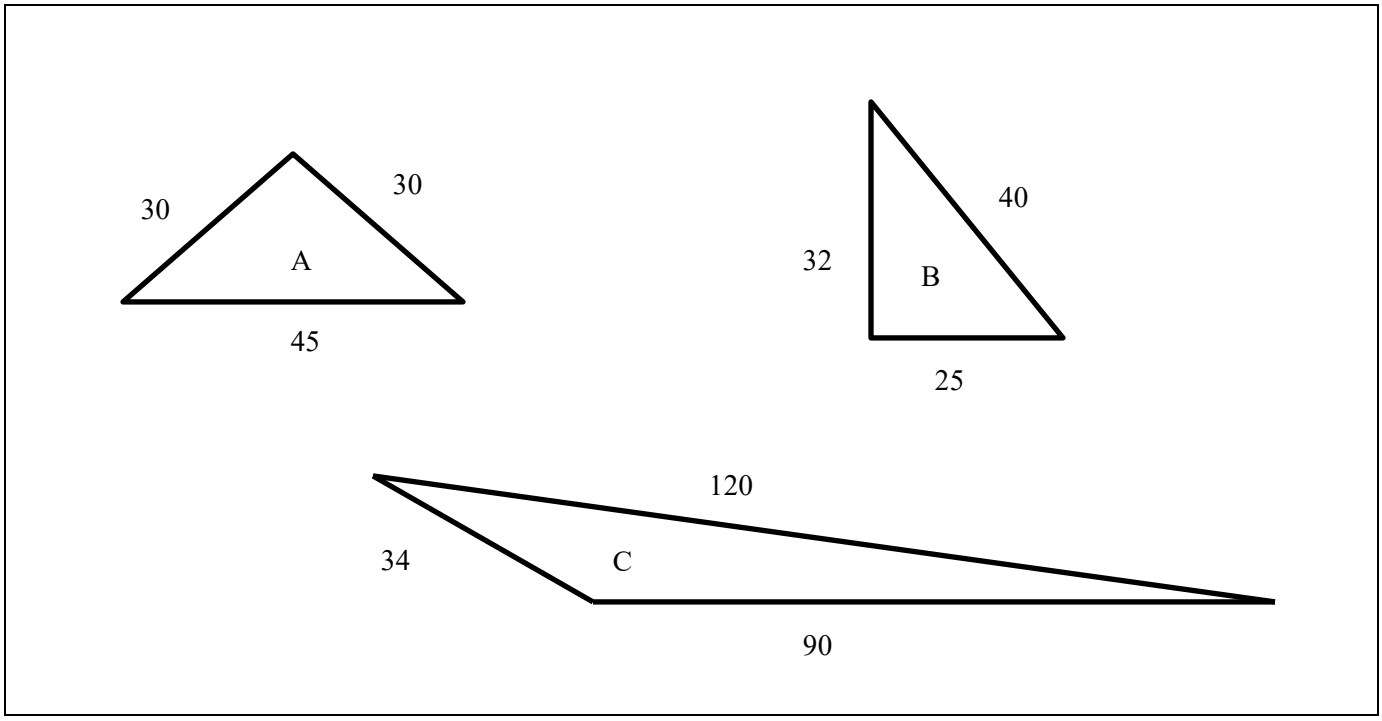
The formula for the area of a triangle, given the three sides is:

$$area = \sqrt{s(s - side1)(s - side2)(s - side3)}$$

where $s = \frac{side1+side2+side3}{2}$

(That is, the value of **`s`** is one-half of the perimeter of the triangle.)

Sample Triangles



Triangle	Side 1	Side 2	Side 3	Perimeter	Area
A	30	30	45	105.00	446.47
B	32	40	25	97.00	399.81
C	34	120	90	244.00	828.92

Code for the toString Method

The **`toString()`** method can be implemented as follows:

```
@Override
  public String toString() {
      return String.format("side1= %6.2f, side2=%6.2f, side3=%6.2f, "
              + "color=%s, filled=%b",
              side1, side2, side3, getColor(), isFilled());
  }
```

Sample Output

The examples that follow show correct output for several test cases. (In these examples, the text that the user types is shown in **BOLD** font. The actual input / output will all be displayed in the same font.)

```
Enter three sides: 3 4 5
Enter the color: Green
Enter a boolean value for filled: false
Triangle: side1=   3.00, side2=  4.00, side3=  5.00, color=Green, filled=false,
Perimeter =  12.00, Area =   6.00
```

```
Enter three sides: 30 30 45
Enter the color: Red
Enter a boolean value for filled: true
Triangle: side1=  30.00, side2= 30.00, side3= 45.00, color=Red, filled=true,
Perimeter = 105.00, Area = 446.47
```

```
Enter three sides: 32 25 40
Enter the color: Orange
Enter a boolean value for filled: true
Triangle: side1=  32.00, side2= 25.00, side3= 40.00, color=Orange, filled=true,
Perimeter =  97.00, Area = 399.81
```

```
Enter three sides: 34 120 90
Enter the color: Purple
Enter a boolean value for filled: false
Triangle: side1=  34.00, side2=120.00, side3= 90.00, color=Purple, filled=false,
Perimeter = 244.00, Area = 828.92
```

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